

Signhify — Local IDE Development Guide

Continue building Signhify in any local AI-native IDE (Google **Antigravity**, **Cursor**, **Windsurf**, **Kiro**, **Gemini CLI**, **Kilo CLI**, **Claude Code**, **Aider**, **Zed**, or plain **VS Code**) while keeping deployments shipping through **Lovable**. Last updated: June 8, 2026 · Maintained by Piyush Raj Singh · hello@signhify.online

This project is bidirectionally synced with GitHub. Anything you push to the default branch is picked up by Lovable within seconds and becomes part of the next preview / publish. Anything you change in Lovable is committed straight back to GitHub. That is the whole contract — everything below is the practical detail to make it boringly reliable.

1. One-time setup

1.1 Prerequisites

Tool	Min version	Why
Node.js	20.x LTS	Required by Vite 7 + TanStack Start
Bun	1.1+	Package manager + script runner (bun . lockb is source of truth)
Git	2.40+	GitHub sync
GitHub account	—	Bidirectional sync with Lovable
Lovable workspace access	—	To publish

Do not swap Bun for npm/pnpm/yarn. The bun . lockb file is the source of truth and Lovable's GitHub sync expects it. Mixing package managers produces lockfile drift and breaks the auto-deploy.

1.2 Connect Lovable ↔ GitHub (do this once)

In Lovable: **Plus (+) menu** → **GitHub** → **Connect project** → **Create Repository**.
Pick the org/account you want. Lovable creates the repo and pushes the current codebase.

1.3 Clone locally

```
git clone git@github.com:<your-org>/signhify.git
cd signhify
bun install
cp .env.example .env # if present, otherwise create .env (see §2)
bun run dev
```

The dev server runs on `http://localhost:3000`. Hot reload, SSR, and TanStack

Router file-based routing all work the same as inside Lovable.

2. Environment variables

Two scopes. Never mix them.

Browser / build-time (VITE_*)

Safe to bundle. Put in local `.env`:

```
VITE_SUPABASE_URL=https://<project>.supabase.co
VITE_SUPABASE_PUBLISHABLE_KEY=eyJ...
```

Server / runtime (process.env.*)

Secrets. Read inside `handler()` of server functions only.

```
SUPABASE_URL=https://<project>.supabase.co
SUPABASE_PUBLISHABLE_KEY=eyJ...
SUPABASE_SERVICE_ROLE_KEY=eyJ...    # never commit, never expose to client
LOVABLE_API_KEY=...                  # optional, for AI gateway
```

On Lovable Cloud the runtime secrets are injected automatically — your local `.env` is only for `bun run dev` on your machine. To rotate or add a runtime secret used in production, do it from the **Lovable** → **Cloud** → **Secrets** panel (not from GitHub).

3. IDE-specific setup

Pick whichever you use. All of them work because they only need (a) the filesystem and (b) `git`.

3.1 Cursor / Windsurf / Zed / VS Code

Open the cloned folder. Recommended extensions:

- Biome / Prettier (formatting already configured via `.prettierrc`)
- ESLint
- Tailwind CSS IntelliSense
- TanStack Router (optional — gives route-tree hints)

Cursor/Windsurf agents: point them at `CONTRIBUTING.md` and `DEPLOY.md` as "always-attached" context so they respect the project's conventions.

3.2 Google Antigravity

1. **File** → **Open Workspace** on the cloned repo.
2. In the Agent panel, add `CONTRIBUTING.md`, `DEPLOY.md`, and this guide to *Knowledge*.
3. Configure the **Run Task** integration to use `bun run dev` and `bun run prepublish:check`.
4. AntigraVity's Browser Use mode targets `http://localhost:3000` — keep the dev server running while you iterate.

3.3 Kiro

1. `kiro init` inside the repo to register the spec workspace.
2. Create a spec at `.kiro/specs/feature-name/` for each new feature; Kiro will scaffold and edit routes under `src/routes/` following the TanStack Start conventions documented in `src/routes/README.md`.
3. Run `kiro hooks add prepublish` so Kiro runs `bun run prepublish:check` before any push to `main`.

3.4 Gemini CLI

```
npm install -g @google/gemini-cli
gemini auth login
cd signhify
gemini                    # interactive session
# or one-shot
gemini -p "add a pricing FAQ section to /pricing"
```

Add a `GEMINI.md` at the repo root if you want to override the default context window. The CLI reads `CONTRIBUTING.md` automatically.

3.5 Kilo CLI

```
brew install kilo-cli      # or curl install per docs
kilo init
kilo run "implement <task>"
```

Kilo respects `.kilo/config.yaml`. Add `prepublish:check` as the pre-commit guard so failed Playwright runs block the commit.

3.6 Claude Code / Aider / OpenCode

Drop in at the repo root and go. They all use `git` for change tracking and will respect `bun.lockb`. For Aider:

```
aider --model claude-3-5-sonnet --read CONTRIBUTING.md DEPLOY.md
```

4. Day-to-day workflow



4.1 Branching

- Work on short-lived feature branches: `feat/<scope>` or `fix/<scope>`.
- Open a PR into `main`. Lovable previews track `main` by default; enable

GitHub Branch Switching in **Account** → **Labs** if you need per-branch previews.

- Squash-merge to keep the Lovable timeline readable.

4.2 Things you must never hand-edit

These are regenerated and will cause merge conflicts or runtime breakage:

- `src/routeTree.gen.ts`
- `src/integrations/supabase/types.ts`
- `src/integrations/supabase/client.ts`, `client.server.ts`, `auth-middleware.ts`, `auth-attacher.ts`
- `bun.lockb` (let `bun install` manage it)

4.3 Database changes

Always go through a migration file:

```
# create the file
touch supabase/migrations/$(date +%Y%m%d%H%M%S)_<name>.sql
# write SQL – remember GRANTS for any new public-schema table
```

Lovable picks up new migrations automatically on deploy. Never run ad-hoc SQL against production from your local shell.

4.4 Server functions vs. server routes

- `createServerFn` (TanStack Start) for internal RPC — files in `src/lib/*.functions.ts`.
- `createFileRoute` with a `server` block for webhooks/public APIs — files in `src/routes/api/public/*`.

Do **not** add Supabase Edge Functions for new work — they're being phased out in favour of TanStack server functions.

5. Pre-publish checklist (mandatory)

Codified in `CONTRIBUTING.md` §8. The short version:

```
bun run prepublish:check
```

This runs the Playwright smoke + HTML-diff suite against the latest preview. It must pass before you publish from Lovable. If you publish without it, the `/publish` audit row will not be created and the gate will block subsequent ships.

Manual gates in addition to the script:

- ☐ Preview health: `/api/public/health` returns `{ok:true}`
- ☐ Marketplace diff is intentional (compare against `src/lib/marketplace.ts`)
- ☐ SEO: every changed route has unique title, description, OG + Twitter tags
- ☐ Security scan clean (no unresolved critical findings)
- ☐ Sync hygiene: `git status` clean, `bun.lockb` unchanged unless deps changed

6. Publishing

You have two equivalent paths:

1. **From Lovable** — open the project, click **Publish** (top-right on desktop, bottom-right on mobile in preview). Frontend changes deploy on click; backend changes (migrations, server functions) deploy automatically as soon as they land on `main`.

2. **From your IDE** — push to `main`, then ask the user (or yourself in Lovable chat) to publish. The agent walks through the Pre-Publish checklist and only then calls the publish tool.

Stable URLs:

Purpose	URL
Production (Lovable)	<code>https://signhify.lovable.app</code>
Production (custom)	<code>https://signhify.online</code>
Preview (stable)	<code>https://project--<project-id>-dev.lovable.app</code>
Production (stable)	<code>https://project--<project-id>.lovable.app</code>

Use the stable `project - -...` URLs for webhooks and cron — they survive project renames.

7. Self-hosting elsewhere (optional)

Even after wiring local dev, you can still host the build outside Lovable. The repo targets Cloudflare Workers / Vercel Edge / Netlify without modification. See `DEPLOY.md` for variables and adapters.

8. Troubleshooting

Symptom	Likely cause	Fix
Failed to resolve import "@/..." after git pull	New file referenced before bun install	Run bun install
Missing Supabase env var(s) in console	.env not loaded	Restart bun run dev; verify variable names
Unauthorized from a createServerFn in SSR	Protected fn called from a public-route loader	Call it from the component via useServerFn (see CONTRIBUTING.md)
Lovable preview shows old code	Push didn't land on main	Check GitHub Actions; force a re-sync from Lovable's GitHub panel
bun.lockb conflict	Two contributors edited deps in parallel	Delete bun.lockb, re-run bun install, commit
Playwright smoke fails locally	Wrong base URL	PLAYWRIGHT_BASE_URL=https://id-preview- <id>.lovable.app bun run prepublish:check</id>
__dirname is not defined in production only	Used a Node-only package in a server fn	Replace with a Workers-compatible alternative (see server-runtime docs)

9. Golden rules

1. **GitHub is the source of truth.** Both Lovable and your IDE are editors on top of it.
2. **Never hand-edit generated files.** They will be overwritten and merge-conflict at the worst moment.

3. **Secrets live in Lovable Cloud, not in `.env` for production.**

4. **Pre-publish check is non-optional.** It exists because we have shipped broken marketplace diffs before.

5. **Small PRs, fast merges.** The Lovable preview rebuild is fast — use it as a real review surface.

Welcome to the codebase. Ship something.